Faisal Mahmud Abir

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LinkedIn | GitHub | Portfolio



PROFILE SUMMARY

Computer Science and Engineering graduate with research experience in natural language processing, computer vision, and efficient deep learning. Developed knowledge-distillation and self-supervised learning pipelines for Bengali hate speech detection and gastrointestinal image classification. Skilled in Python, PyTorch, and Hugging Face Transformers, with hands-on experience in model development, fine-tuning, and evaluation. Seeking an AI/ML research role focused on efficient models and representation learning.

TECHNICAL SKILLS

Programming & Data: Python, C, C++, JavaScript, SQL, NumPy, Pandas, Advanced Excel, Power BI

AI & Machine Learning: PyTorch, Scikit-Learn, Hugging Face Transformers, YOLO, OpenCV; machine learning, deep learning, NLP, computer vision, generative AI, RAG, knowledge distillation, self-supervised learning, fine-tuning

Backend, Web & Databases: FastAPI, Flask, REST APIs, WebSockets, React.js, CSS, PostgreSQL, MongoDB, ChromaDB

Cloud, Deployment & Automation: AWS, GCP, Docker, MLOps, Git, GitHub, Linux, n8n

RESEARCH EXPERIENCE

Distilling Hate: Lightweight Transformer Models for Fine-Grained Bengali Hate Speech Detection | GitHub

Undergraduate Capstone Thesis | *Manuscript Under Review at PLOS ONE journal*

• Developed a lightweight Bengali hate speech detection model by distilling BanglaBERT Large (336.7M parameters) into a 13.7M-parameter student using cross-entropy, KL divergence, and hidden-state alignment.

• Reduced inference FLOPs by 96.8% and evaluated performance on BanglaMultiHate and BanTH, achieving a macro-F1 of 0.5799.

JSSL-GI: Joint Self-Supervised Learning for Multi-Category Gastrointestinal Endoscopy Image Classification | GitHub

Manuscript Under Review at IEEE Access

• Developed a joint self-supervised learning pipeline using an ImageNet-pretrained ViT-Small, combining masked autoencoder reconstruction and multi-view contrastive learning with adaptive uncertainty-based loss weighting.

• Evaluated the model across 23 Hyper-Kvasir classes, achieving 92.00% accuracy and 67.30% macro-F1 after 30 pretraining epochs.

EDUCATION

B.Sc. in Computer Science and Engineering — Spring 2022 – Spring 2026
North South University — CGPA: 3.08/4.00

Higher Secondary Certificate — 2020 – 2021
Chittagong University College — GPA: 5.00

Secondary School Certificate — 2017 – 2018
Hathazari Parboti Govt. High School — GPA: 5.00

PROJECTS

Agentic Safety & Security Surveillance System | GitHub | Live Demo Aug 2026 - Sep 2026

• Built a FastAPI and React video-analysis module using YOLO11s and ByteTrack for person detection and tracking, with annotated videos, event summaries, and email alerts for model-flagged events.

Bangla Voice Assistant for Online Forms | GitHub Jun 2026 - Jul 2026

• Developed a Chrome extension with Bangla & English voice input, LLM-based translation, and automated form filling; implemented field validation, read-back verification, undo, session recovery, and user-confirmed submission.

MemeGuard AI | GitHub | Live Demo Jan 2026 - Apr 2026

• Built a real-time multimodal hate-speech moderation platform using ResNet-18 and DistilBERT, with Flask, REST APIs, and WebSockets for web and Telegram integration.

MultiDoc AI | GitHub Jul 2025 - Oct 2025

• Built a RAG document question-answering system supporting PDF, DOCX, CSV, and XLSX; integrated ChromaDB semantic retrieval with FastAPI for context-aware responses.

ACHIEVEMENTS & LEADERSHIP

Lead Coordinator — IEEE NSU IAS Student Branch Chapter 2023 - 2025

• Led 24+ technical and professional development programs, coordinating logistics, budgets, speakers, sponsors, and faculty as part of the Events & Management Team.

• Received the Best Lead Award (2024–2025) for leadership and team coordination.